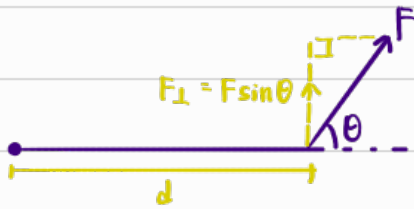


## Moment / Torque

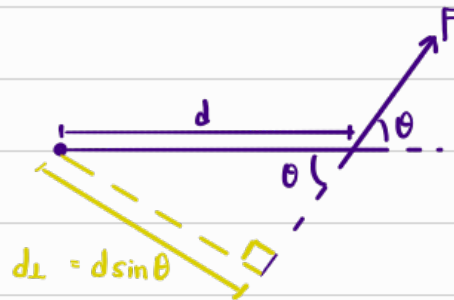
$$\tau = F_{\perp} \times d \quad \text{OR} \quad F \times \perp d$$

$$= (F \sin \theta)(d) \quad \text{OR} \quad F(d \sin \theta)$$

Moment arm [Perp. dist.]



OR

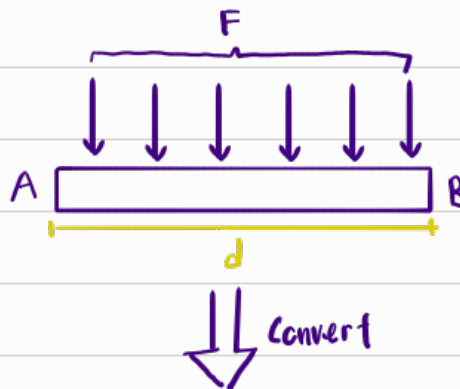


## Idealised loads

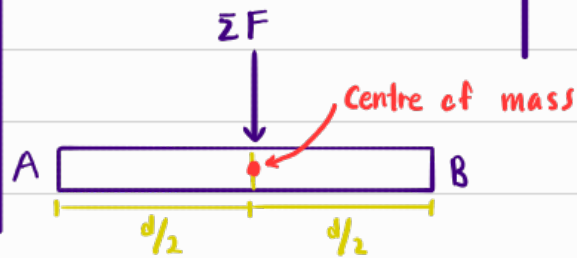
• Concentrated force



• Distributed load



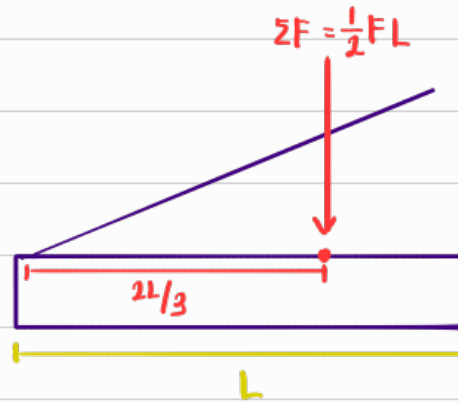
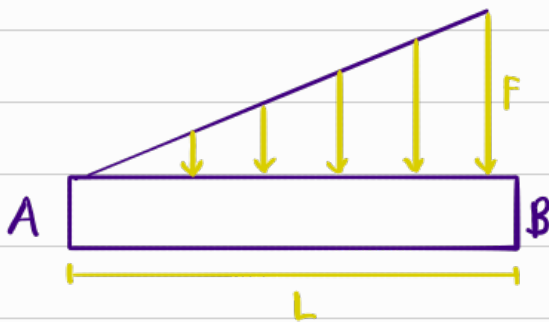
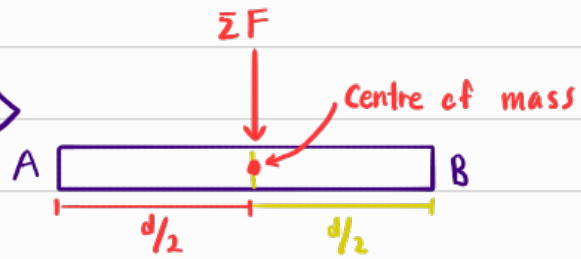
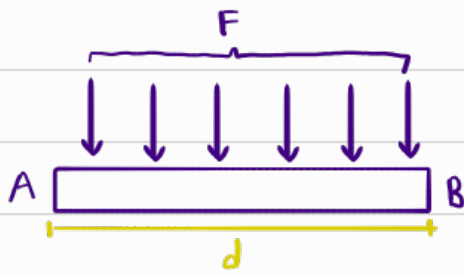
Concentrated Force



• Applied moment (Nm)

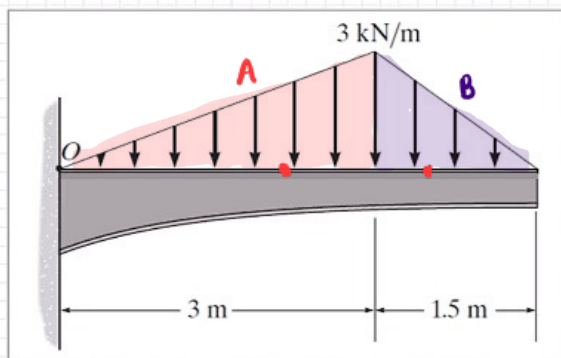


## Converting uniform $\rightarrow$ Concentrated Force



## Example

Consider the loading below. Replace the distributed loading with the equivalent point forces and specify its locations on the beam measured from point O.

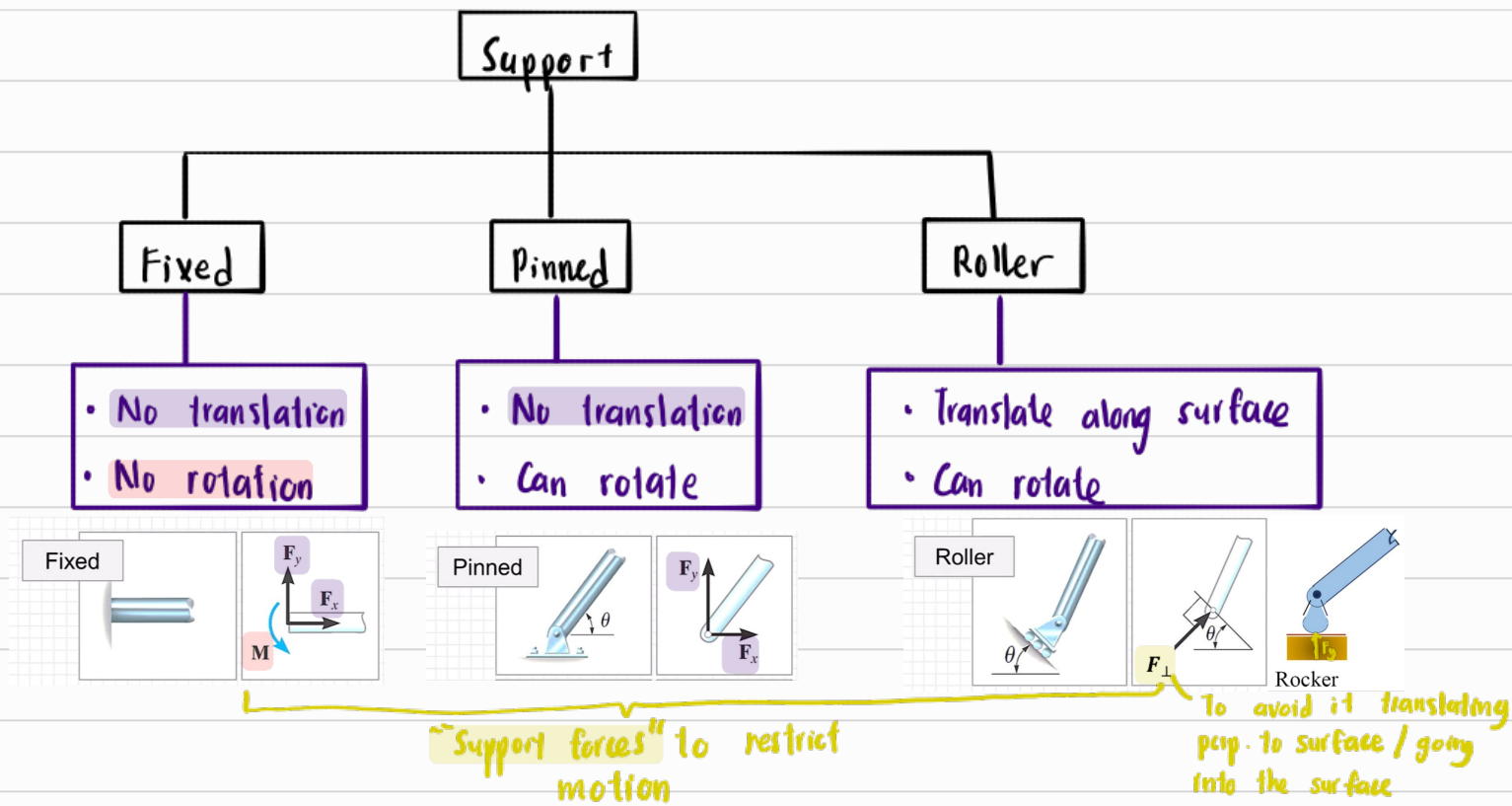


$$\bar{a}_x = \frac{2(3)}{3} = 2 \text{ m from O}$$

$$\Sigma F_A = \frac{1}{2}(3)(3) = 4.5 \text{ kN}$$

$$\bar{b}_x = 3 + \frac{1.5}{2} = 3.5 \text{ m from O}$$

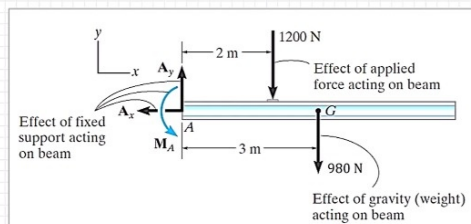
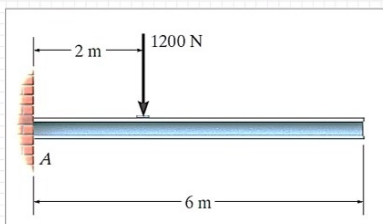
$$\Sigma F_B = \frac{1}{2}(3)(1.5) = 2.25 \text{ kN}$$



## Example 1 :

Draw a free-body diagram of the 100 kg cantilever.

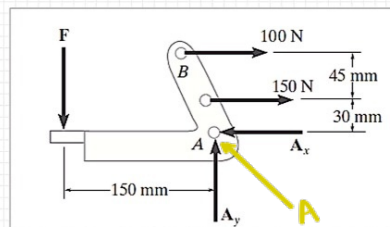
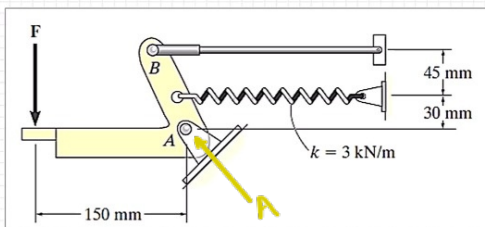
- Fixed support (3 unknown reactions)
- 1200 N external force
- Weight of the beam



## Example 2

Draw a free-body diagram of the foot lever. The spring is stretched 50 mm and the force on link B is 100 N.

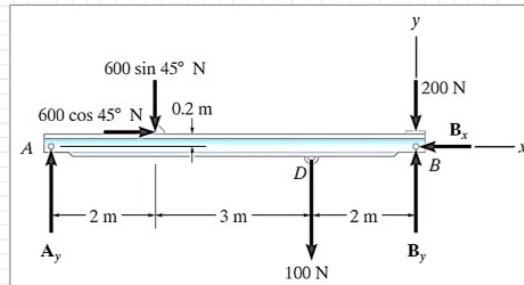
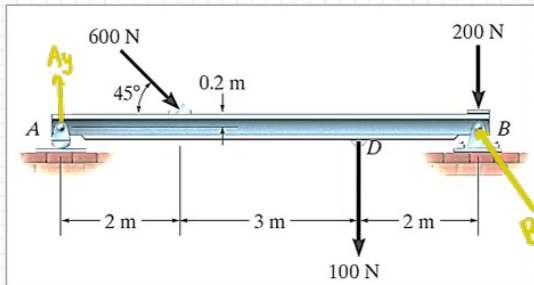
- Pin support at point A (2 unknown reactions)
- Weightless link at point B (1 reaction)
- Various external forces



### Example 3

Draw a free-body diagram of the beam.

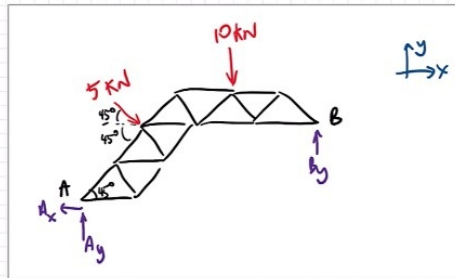
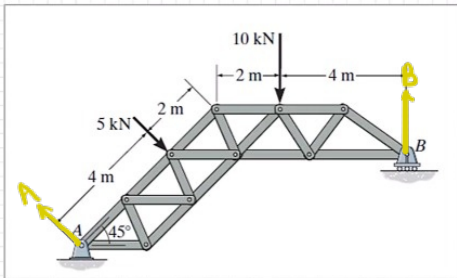
- Rocker support at point  $A$  (1 unknown reaction)
- Pin support at point  $B$  (2 unknown reactions)
- Various external forces



### Example 4

Draw a free-body diagram of the truss.

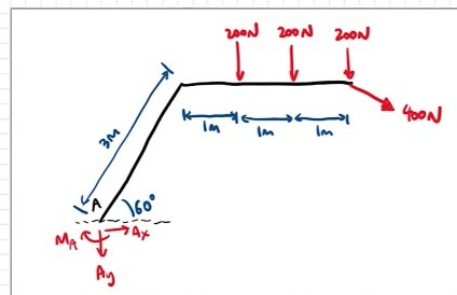
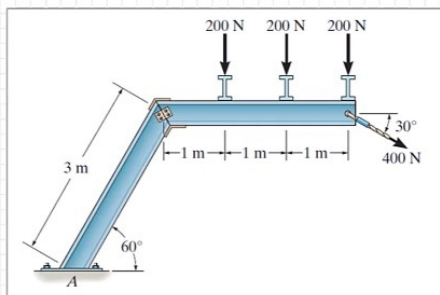
- Pin support at point  $A$  (2 unknown reactions)
- Roller support at point  $B$  (1 unknown reaction)
- Various external forces



### Example 5

Draw a free-body diagram of the structure.

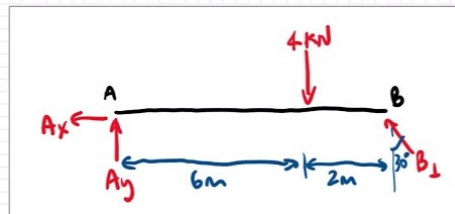
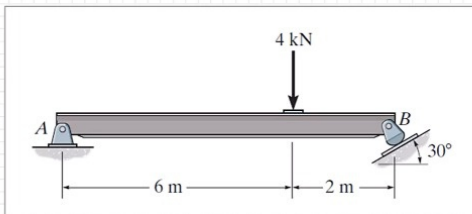
- Fixed support at point  $A$  (3 unknown reactions)
- Cable force in the hole (1 reaction)
- Various external forces



## Example 6

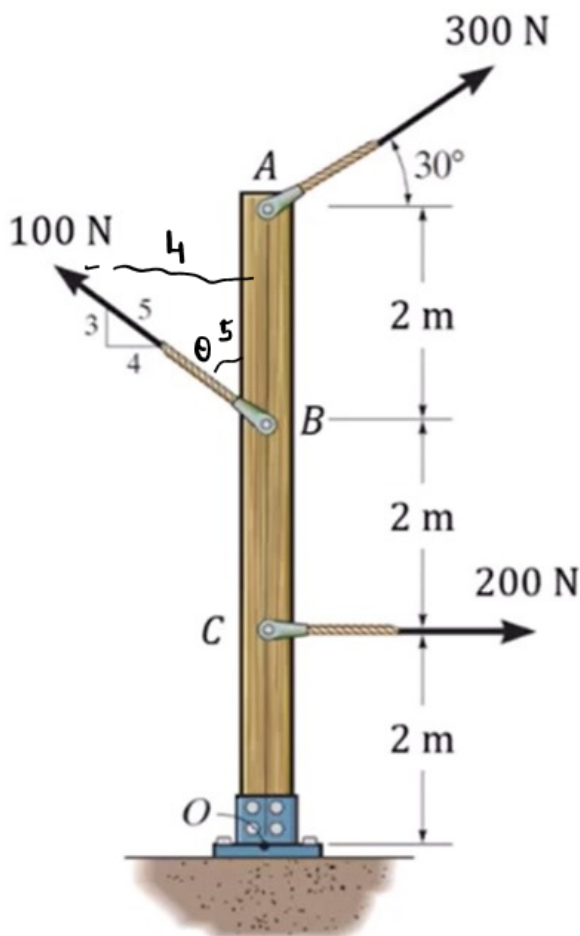
Draw a free-body diagram of the beam.

- Pin support at point  $A$  (2 unknown reactions)
- Rocker support at point  $B$  (1 unknown reactions)
- 4 kN external force



## Example 1

Consider the vertical beam fixed to the ground at point  $O$ . Determine the **resultant** moment about points  $A$ ,  $B$ ,  $C$ , and  $O$  for the system below. Note this is asking for the resultant moment in the system about a specific point, not the **reaction** moment of the fixed support.



About A

$$\tau_{cw} = 100 \left( \frac{4}{5} \right) (2)$$

$$\tau_{Acw} = 200 (4)$$

$$\therefore \Sigma \tau = 200 (4) - 100 \left( \frac{4}{5} \right) (2)$$

$$= 640 \text{ Nm} \#$$

About point B

$$\tau_{cw} = (300 \cos 30^\circ) (2)$$

$$\tau_{Acw} = 200 (2)$$

$$\therefore \Sigma \tau = 200 (2) - 300 \cos 30^\circ (2)$$

$$= -119.6152 \text{ Nm} \#$$

About C

$$\tau_{cw} = (300 \cos 30^\circ) (4)$$

$$\tau_{Acw} = 100 \left( \frac{4}{5} \right) (2)$$

$$\therefore \Sigma \tau = \tau_{Acw} - \tau_{cw}$$

$$= -879.2305 \text{ Nm} \#$$

About O

$$\tau_{cw} = (300 \cos 30^\circ) (6) + 200 (2)$$

$$\tau_{Acw} = 100 \left( \frac{4}{5} \right) (4)$$

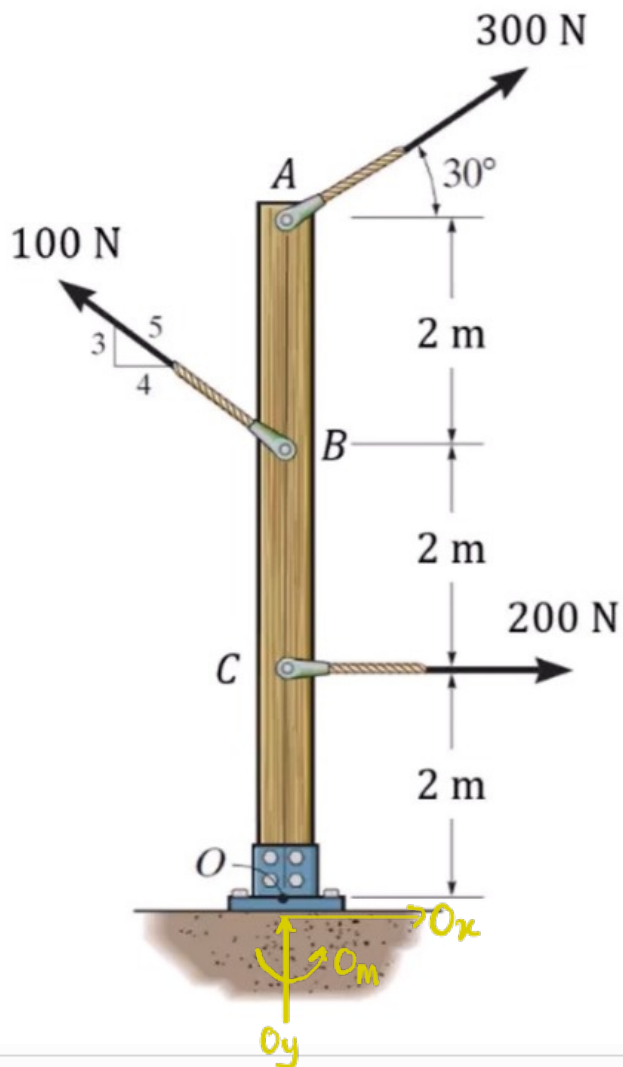
$$\Sigma \tau = \tau_{Acw} - \tau_{cw} = -1638.8457 \text{ Nm} \#$$

$\curvearrowright$  +ve



## Example 2

Consider the vertical beam fixed to the ground at point  $O$ . Determine the reactions of the fixed support.



$$\sum F_x = 0$$

$$O_x + 200 - 100\left(\frac{4}{5}\right) + 300 \cos 30^\circ = 0$$

$$O_x = -379.8076\text{ N}$$

$$\sum F_y = 0$$

$$O_y + 100\left(\frac{3}{5}\right) + 300 \sin 30^\circ = 0$$

$$O_y = -210\text{ N}$$

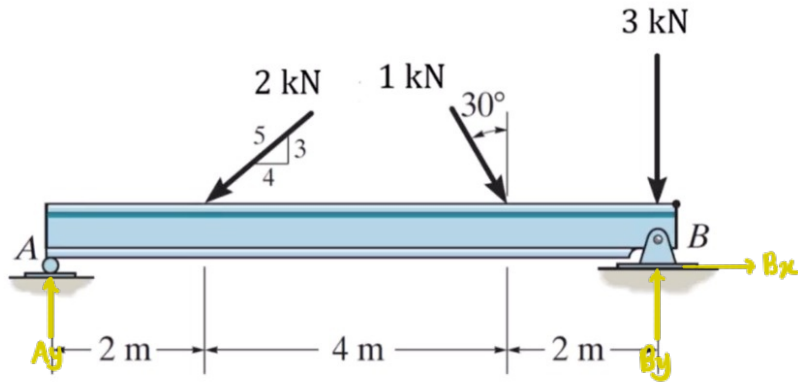
From Example 1,  $\sum \mathcal{M}$  at  $O$  is  $-1638.8457\text{ Nm}$ .

$$\therefore O_m = 1638.8457\text{ Nm}$$

### Example 3



Consider the beam below which is supported by a roller support at  $A$  and a pin support at  $B$ . Determine the reaction forces at the supports. Assume the thickness of the beam to be negligible.



$$\sum F_x = 0$$

$$-2\left(\frac{4}{5}\right) + 1\sin 30^\circ + B_x = 0$$

$$B_x = 1.1 \text{ kN}$$

$$\sum F_y = 0$$

$$A_y - 2\left(\frac{3}{5}\right) - 1\cos 30^\circ - 3 + B_y = 0$$

$$A_y = 1.1165 \text{ kN}$$

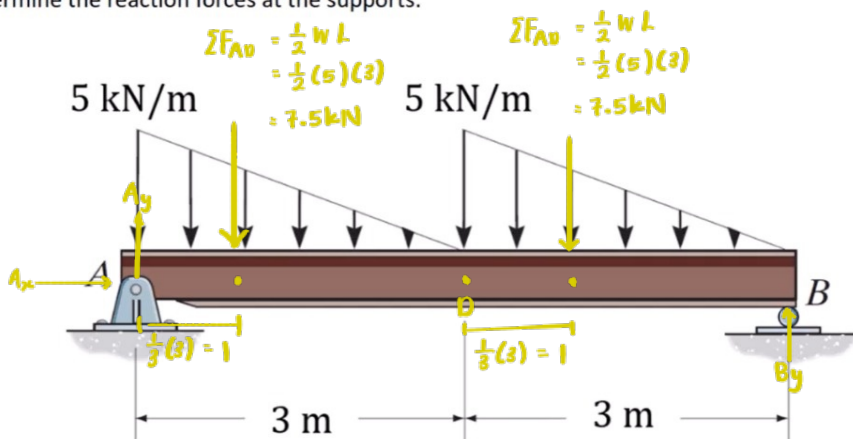
$$\sum \mathcal{L}_A = 0$$

$$-2\left(\frac{3}{5}\right)(2) - 1\cos 30^\circ(6) - 3(8) + B_y(8) = 0$$

$$B_y = 3.9495 \text{ kN}$$

### Example 4

Consider the beam below, which is supported by a pin support at  $A$  and a roller support at  $B$ . Determine the reaction forces at the supports.



$$\sum F_x = 0$$

$$A_x = 0$$

$$\sum \mathcal{L}_A = 0$$

$$-7.5(1) - 7.5(4) + B_y(6) = 0$$

$$B_y = 6.25 \text{ kN}$$

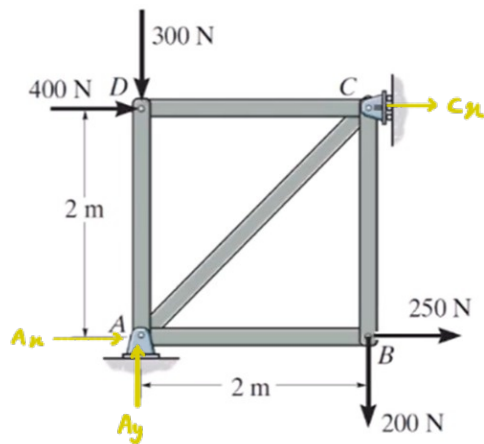
$$\sum F_y = 0$$

$$A_y - 7.5 - 7.5 + B_y = 0$$

$$A_y = 8.75 \text{ kN}$$

### Example 5

Consider the truss below which is supported on a pin joint at point A and a roller joint at point C. Determine the reaction forces at the supports.



$$\sum F_y = 0$$

$$A_y - (300 + 200) = 0$$

$$A_y = 500 \text{ N}$$

$$\sum F_x = 0$$

$$400 + C_x + A_x + 250 = 0$$

$$A_x = -50 \text{ N}$$

$$\sum \tau_A = 0$$

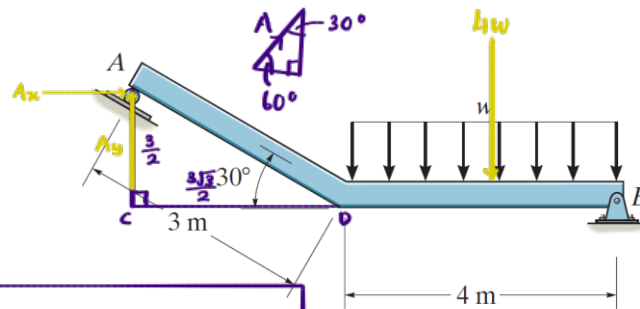
$$-400(2) - C_x(2) - 200(2) = 0$$

$$C_x = -600 \text{ N}$$



### WORKSHOP QUESTION 3

Consider the following beam. It is designed to hold a distributed load of  $w$  N/m on the horizontal portion of the beam. The roller at point A can support a maximum force magnitude of 4 kN and the pin at B can support a maximum force magnitude of 8 kN? Use the problem-solving procedure to work through this question. Detail each stage of the problem-solving procedure. What is the max value of  $w$ ?



$$A \leq 4 ; B \leq 8$$

$$\sin 30^\circ = \frac{AC}{3} \Rightarrow AC = \frac{3}{2}$$

$$\cos 30^\circ = \frac{CD}{3} \Rightarrow CD = \frac{3\sqrt{3}}{2}$$

$$\sum \tau_B = 0$$

$$(A \cos 60^\circ) \left(\frac{3}{2}\right) + (A \sin 60^\circ) \left(4 + \frac{3\sqrt{3}}{2}\right) - 4w(2) = 0$$

$$A = 1.1376w \quad \text{--- ①}$$

$$\sum F_x = 0$$

$$A \cos 60^\circ - B_x = 0$$

$$B_x = 0.618802w, \text{ [using ①]}$$

$$\sum F_y = 0$$

$$A \sin 60^\circ + B_y - 4w = 0$$

$$B_y = 2.9181w, \text{ [using ①]}$$

$$B = \sqrt{B_x^2 + B_y^2}$$

$$= 2.9919w$$

|                                                                 |                                                       |
|-----------------------------------------------------------------|-------------------------------------------------------|
| $\therefore A < 4$<br>$1.1376w < 4$<br>$w < 3.231 \text{ kN/m}$ | $B < 8$<br>$2.9919w < 8$<br>$w < 2.6730 \text{ kN/m}$ |
| $\therefore w < 2.6730 \text{ kN/m}$                            |                                                       |